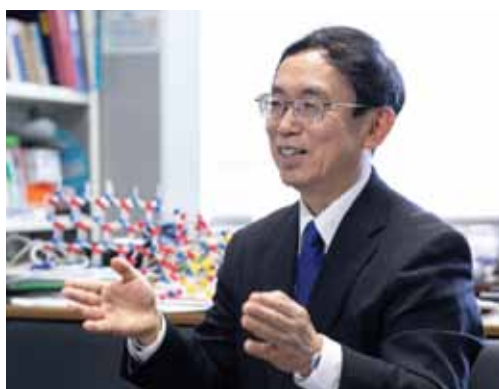
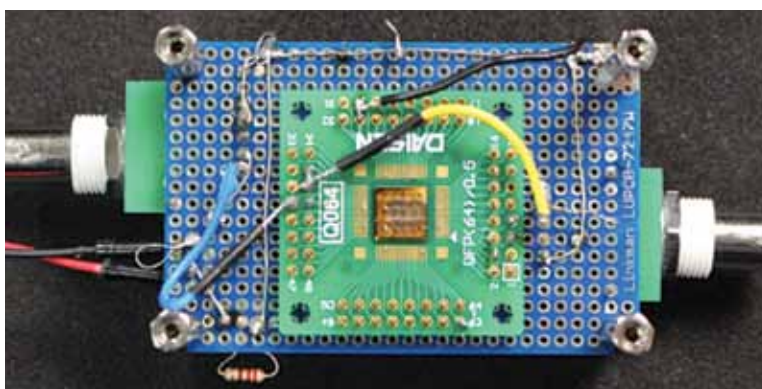


DIAMOND SEMICONDUCTORS: The Key To Next-Generation Power Electronics

With unmatched thermal conductivity and a wide bandgap, diamonds are poised to meet the increasing demand for more efficient, high-performance devices.



Makoto Kasu



Industrial diamond in the base of a semiconductor

(Source: Saga University)



DR KRISHNA NAGA
works as a Physics teacher. She has over twenty years of experience in this field. Her hobbies include reading and writing

Diamonds, long symbols of love, power, and wealth, are formed in Earth's mantle and are prized for their exceptional hardness, brilliance, and rarity. These qualities not only make them valuable as gemstones but also essential in various industrial applications, including cutting tools, abrasives, heat sinks, optical components, drilling, and machinery. Their high refractive index and transparency further enhance their value in high-performance lenses and laser windows.

Beyond their aesthetic appeal, diamonds have garnered attention in the semiconductor industry for their unique properties. Lab-grown synthetic diamonds, produced primarily through microwave plasma chemical vapour deposition (MPCVD), share the same

chemical and physical characteristics as natural diamonds, while addressing many ethical and environmental concerns.

The potential of diamonds to surpass materials like silicon carbide (SiC) and gallium nitride (GaN) in semiconductor applications has driven significant research worldwide. Universities and research institutions in the US, Europe, and Asia are at the forefront of this promising development.

First diamond n-channel MOSFET

The first power circuit containing synthetic diamond semiconductors has been developed by a Japanese university research team. After exploring the theory that diamond semiconductors can outperform silicon and other materials currently used, the team at Saga University, led by